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Code

JupyterLab | Python 3 (ipykernel)

[31]:
```
import pandas as pd
import numpy as np
np.random.seed(42)
email = np.array([10,20,30,40,55,68,70,90]).reshape([-1,1])
spam_email = np.array([0,0,1,1,0,1,0,1]).reshape([-1,1])
print(email.flatten())
print(spam_email.flatten())
```

```
[10 20 30 40 55 68 70 90]
[0 0 1 1 0 1 0 1]
```

[32]:
```
data = pd.DataFrame({"email" :email.flatten(),"spam_email":spam_email.flatten()})
print(data)
```

```
   email  spam_email
0     10           0
1     20           0
2     30           1
3     40           1
4     55           0
5     68           1
6     70           0
7     90           1
```

[33]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(email,spam_email,test_size=0.3,random_state=42)
print('training:',(len(x_train)))
print(f"testing: {len(x_test)}")
```

```
training: 5
testing: 3
```

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Code | JupyterLab | Python 3 (ipykernel)

```
testing: 3
```

[48]:

```
from sklearn.linear_model import LogisticRegression
model=LogisticRegression()
model.fit(x_train,y_train.flatten())
slope = model.coef_[0][0]
intercept = model.intercept_[0]
print("slope:",(slope))
print("intercept:",(intercept))
```

```
slope: -0.020383892533488226
intercept: 1.5844945006808828
```

[50]:

```
y_pred = model.predict(x_test)
y_pred_proba = model.predict_proba(x_test)
new_email_length = np.array([[500]])
prediction = model.predict(new_email_length)[0]
probability = model.predict_proba(new_email_length)[0]
print(f"  Prediction: {'SPAM' if prediction == 1 else 'NOT SPAM'}")
print(f"  Confidence: {max(probability)*100:.1f}%")
```

```
Email with 500 characters:
  Prediction: NOT SPAM
  Confidence: 100.0%
```

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